

GROUP NUMBER: 70

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TEST: The goal is to test the performance of **Sticky Sampling (SS)** and **Count-Min (CM)** for different input configurations. We refer to an item as *frequent/almost frequent/rare* if its true frequency is in the interval $[n\phi, +\infty) / [n(\phi - \epsilon), n\phi), [0, n(\phi - \epsilon))$, respectively.

You must fill in the following tables using the input configurations indicated in red for each table. You must run each test three times and report the average values of the three runs

STICKY SAMPLING: $n=1000000$, $\phi=0.07$, $\delta=0.05$, port=8888				
ϵ	Number of frequent items returned by SS	Number of almost frequent items returned by SS	Number of rare items returned by SS	Number of elements stored in the dictionary used by SS
0.01	10.00	0.00	0.00	41.33
0.02	10.00	0.00	0.00	26.33
0.04	10.00	2.67	0.00	19.67

COUNT MIN: $n=1000000$, $\phi=0.07$, $d=5$, port=8888				
w	Number of frequent items returned by CM	Number of almost frequent items (w.r.t. $\epsilon=0.04$) returned by CM	Number of rare items (w.r.t. $\epsilon=0.04$) returned by CM	Total number of items returned by CM (sum of previous 3 columns)
15	10.00	0.33	232.67	243.00
30	10.00	0.33	14.33	24.67
60	10.00	0.00	2.00	12.00

ANSWER THE FOLLOWING QUESTIONS

- Did you use some Generative AI tool? YES
- We used Claude AI and Gemini. On the code side, they were mainly used to understand and implement the Spark-specific parts (from the template program), to assist with the translation of some of the math to java and for a final check of correctness and code quality. Beyond that, they were used to discuss the meaning of the results and to thoroughly understand how the theory translated into practice.